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Key Points:

- Analysis of Uranus' Lyman α emissions from 1998 to 2022 require a decreasing H scale height after 1998 corresponding to the H_3^+ cooling trend
- The exospheric cooling since 1998 is unaffected by auroral activity and is not emulated at altitudes below the 1 μ bar pressure level
- This indicates a delay in the response of the exosphere to the thermospheric cooling trend of the H_3^+ emissions

Supporting Information:

Supporting Information may be found in the online version of this article.

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Steady Collapse of Uranus' Exosphere After 1998 to the Present Decade

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Abstract Uranus' thermospheric temperature decreased from ~ 800 K in 1986 to ~ 450 K in 2022 as determined from observations of H_3^+ and H_2 infrared emissions. Spitzer 2007 lower atmosphere observations do not emulate this cooling trend. Here we show that the atomic H Lyman α emission from the disk of Uranus observed by HST from 2011 to 2022 are not consistent with radiative transfer models based on a constant atmospheric structure retrieved from the Voyager 2 flyby of 1986. Instead, the optical depth of the H column matching the Uranus Lyman α disk brightness decreased after 1998, consistent with the long-term cooling trend. This decrease is irrespective of auroral activity. While the origin of the cooling is poorly understood, it indicates that the density and extent of the Uranian exosphere changes on a time scale of years impacting the atmospheric structure, the magnetospheric proton source, and exospheric drag on the inner rings.

Plain Language Summary Uranus' thermosphere has been steadily cooling since the Voyager 2 flyby in 1986 to the present decade. Spitzer spacecraft's infrared observations indicate that this cooling is not matched by the lower atmosphere (pressure $> 1 \mu$ bar). We have used Hubble Space Telescope observations of the upper atmosphere of Uranus in the far ultraviolet from 1998 to 2022 to show that the region above the thermosphere, that is, the exosphere, has also been experiencing long-term cooling like the thermosphere, but only after 1998, irrespective of auroral activity. The processes responsible for this upper atmospheric cooling are unknown at present. Determining the vertical density structure of the thermosphere and exosphere and its correlation with Uranus' seasons and solar wind intensity provides key insight into upper-atmospheric variability. For example, the seasonal changes in upper atmospheric density drive the amount of frictional drag experienced by ring particles, which in turn determines the lifetime and temporal survivability of the ring system. Variations in density also regulate the amount and energy of charged particles entering the upper atmosphere along open magnetic field lines. These particles heat the upper atmosphere and contribute significantly to the planet's total energy budget.

1. Introduction

When Voyager 2 arrived at Uranus in 1986 it encountered an unusual and intriguing planet. Not only did it find that Uranus' rotational axis was almost aligned with the Solar System ecliptic plane but also discovered an offset magnetic field for the planet tilted at an angle of 60° from the rotational axis (Ness et al., 1986). Expanding on Earth-based UV observations (Clarke et al., 1986) the Voyager 2 ultraviolet spectrometer (UVS) detected Lyman α emission from an unexpectedly hot and extended exosphere with a temperature of ~ 800 K and gas densities of $\sim 2 \times 10^3 \text{ cm}^{-3}$ (thermal + non-thermal) at an altitude of $\sim 25,000$ km or 1 Uranian radius (R_U) (Herbert & Hall, 1996). The 800 K temperature was twice as hot as the temperature derived for Saturn's exosphere (Sandel et al., 1982). This high temperature value could not be explained by the UV energy input received by the upper atmosphere of Uranus from the Sun.

Since the Voyager 2 flyby, Uranus has been periodically observed both in the ultraviolet and the infrared using space-based and ground-based observatories. Far-ultraviolet (FUV) observations with the Hubble Space Telescope (HST) detected faint auroral activity possibly correlated with the arrival of interplanetary shocks, triggering large-scale magnetospheric compressions (Lamy, 2020; Lamy et al., 2012, 2017, 2025). This auroral activity was believed to be otherwise inhibited during the equinox season due to the decrease in strength of the bipolar rotating magnetic tail formed by the 60° separation of the magnetic axis from the rotational axis (Cowley, 2013). This

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means that because of the extreme tilt of the magnetic axis, the magnetotail does not stay aligned in the anti-sunward direction. Instead, it sweeps around Uranus bending and weakening the current system driving the aurora during equinox. Infrared observations of Uranus include disk-integrated emission-line spectra from H_3^+ ions present in the ionosphere and H_2 present in the lower thermosphere (Melin et al., 2019; Trafton & Kaplan, 2025). Both these emissions have been intermittently observed from the early 1990s to the present. Notably both these emissions reveal a steady decrease in ionospheric and thermospheric temperature from ~ 690 K in 1993 to ~ 450 K in 2023 (Trafton & Kaplan, 2025; Trafton et al., 1993). In contrast, the lower and middle atmospheres (pressure > 1 μbar) do not show evidence of a long-term decrease in temperature between 1986 and 2007 (Orton, Fletcher, et al., 2014; Orton, Moses, et al., 2014; Orton et al., 2015).

At present, two hypotheses have been put forth to explain the long-term decrease in the thermospheric and ionospheric temperatures observed in the infrared data. The first one highlights the concurrent decrease in the solar wind dynamic pressure on the Uranian magnetosphere ($\sim 50\%$ from 1992 to 2018) resulting in an expanded structure and weaker electrodynamic heating of the thermosphere, as the cause of the observed long-term cooling trend (Masters et al., 2024). The second one proposes the existence of a magnetic season at Uranus produced by the slow changing geometry of the magnetic field toward the impinging solar wind resulting in variable heating of the upper atmosphere driven by particle precipitation and Joule heating (Melin et al., 2013, 2019). This magnetic season is unique to Uranus due to the 60° tilt separation of the magnetic pole from the rotational axis and the magnetic dipole being offset from the planet's center by $\sim 0.3 R_U$. At present, the primary driver of long-term thermospheric and ionospheric cooling at Uranus is unknown.

A recent analysis of Uranus' H Lyman α emission from 1998 to 2011 obtained with HST concluded that the temperature structure of Uranus' exosphere (pressure < 1 μbar) has remained unchanged between the Voyager flyby in 1986 and 2011 (Joshi et al., 2025), despite the thermospheric temperature dropping from ~ 800 to ~ 550 K during this period. This result and the previous studies about the lower atmosphere imply that the cooling trend observed at Uranus, from H_3^+ and H_2 observations, is confined only to the thermosphere region and does not percolate either into the lower, denser regions of the atmosphere or into the exosphere.

In this paper, we have analyzed HST observations of Lyman α emission from the disk of Uranus from 1998 to 2022, a longer period than the previous analysis of HST Lyman α data (1998–2011). The observations span multiple solar activity cycles and a little less than a third of Uranus' 84-year orbit around the Sun. The analysis reveals the global state of the exosphere of Uranus over a 24-year period when the thermospheric temperature decreased from ~ 650 to ~ 450 K. The result of this study has implications for the upper atmospheric density and altitudinal extent at Uranus as it orbits the Sun, which in turn affects the lifetime of small ring particles through frictional drag and is important for planning the Uranus Orbiter and Probe Mission (Broadfoot et al., 1986; National Academies of Sciences, Engineering, and Medicine, 2023).

1.1. HST Observations of Uranus' Exosphere

Uranus' orbit around the Sun takes ~ 84 years. During the Voyager 2 flyby in 1986 Uranus was experiencing southern summer solstice. Uranus was at equinox around December 2007 and will be at northern summer solstice in April 2030 (Sromovsky et al., 2024; JPL Horizons). The HST data set used in this analysis spans 1998–2022, that is, ~ 10 years before equinox up until ~ 8 years before northern summer solstice (see Figure S1 in Supporting Information S1). Specifically, we have used HST observations with the Space Telescope Imaging Spectrograph (STIS) instrument in the far ultraviolet to analyze the Lyman α emission intensity from the disk of Uranus. The data consists of Multi Anode Microchannel Array images from two different observing modes: (a) STIS + G140 L images wherein the low-resolution G140 L grating is placed in the optical path of the telescope with no spectrograph aperture to obtain an image of Uranus at Lyman α and (b) STIS broadband observations consisting of a clear (FUV-MAMA from 1,150 to 1,800 Å) and a filtered (F25SrF2 from 1,280 to 1800 Å) image of Uranus, the subtraction of which gives a final image containing only Lyman α emission (see Figure S2 in Supporting Information S1). The final reduced Lyman α image is used to construct radial intensity profiles in 90° angular slices from the center of the planet in the images following the same procedure as Clarke et al. (2014) and Bhattacharyya et al. (2015, 2023). This procedure averages pixel intensities equidistant from the planet's center covering a larger radial arc at longer distances thereby improving the signal-to-noise ratio (SNR) away from the planet. The radial intensity profile from each quadrant is then combined to further improve upon the SNR of the Lyman α signal from Uranus for each observation day.

The Lyman α profile of Uranus obtained from the STIS image contains background counts from interplanetary hydrogen (IPH) + Earth's exospheric H atoms, or the geocorona, due to HST being in low-Earth orbit. The geocoronal H Lyman α emission is the largest background contributor and its intensity varies with HST's orbit around Earth from ~ 1 kR at the anti-sunward point on the nightside to 30 kR at the sub-solar point on the dayside (Bhattacharyya et al., 2020; Waldrop & Paxton, 2013). For the entire exposure with STIS this geocoronal Lyman α emission appears as an additive constant background against the Uranus Lyman α profile (see Figure S3 in Supporting Information S1). This additive nature is only possible when the central narrower Uranus H Lyman α line profile ($T < 1,000$ K) is sufficiently Doppler shifted from the geocoronal Lyman α line profile ($T \sim 1,000$ K) otherwise most of the Uranus H Lyman α photons will undergo multiple scattering by the geocoronal H atoms before being detected by HST (~ 500 km altitude), rendering it impossible to determine the true Lyman α flux from Uranus at the top of the Earth's exosphere ($\sim 60 R_E$). We set this Doppler shift threshold to be greater than ± 10 km/s for the observations (see Figure S4 in Supporting Information S1). We estimate this background constant value (IPH + geocorona) from the intensity profile $> 1.7 R_U$ and subtract it from the entire profile to eliminate the background Lyman α emission sources. This results in a slight over subtraction from the Uranus disk Lyman α signal as the planet's disk blocks IPH emissions from Uranus to infinity whereas the limb does not. Because we cannot separate the IPH signal from the geocorona, we correct for the IPH over-subtraction during the modeling process by carefully considering the IPH column density between Uranus and infinity along the Earth-Uranus direction vector using an IPH model (Quémerais et al., 2013) and adding that value back to the Uranus Lyman α disk intensity in the data before modeling. The modeling also accounts for scattering by the IPH and geocoronal column between Uranus and Earth where the final observation is taken with HST. Point spread function effects are not considered in the modeling as it is restricted to the disk where such distortions are negligible. Because the observations in each year are close in time, the profiles for each year have been averaged to obtain a single profile with an improved SNR. The uncertainty in the profile is calculated following standard error propagation techniques starting from Poisson statistics being applied to the raw image counts. Table S1 in Supporting Information S1 provides a summary of the HST observations, and the related parameters used in this work.

2. Modeling the Uranian Exospheric Lyman α Emission

The Voyager 2 FUV spectra of Uranus revealed a Raman line at $1,280 \text{ \AA}$ not seen at Jupiter, Saturn, or Neptune (Yelle et al., 1987). Its presence indicates that at Uranus hydrocarbons that are able to absorb Lyman α photons are absent above the 0.5 mbar pressure level. This is because the homopause is much deeper on Uranus than on the other giant planets (Herbert et al., 1987; Orton, Fletcher, et al., 2014; Orton, Moses, et al., 2014) resulting in significant scattering of Lyman α by H_2 in altitudes just above the homopause. Therefore, unlike the other giant planets in our Solar System, modeling the Uranus disk Lyman α emission requires three processes: resonant scattering of solar Lyman α by atomic H, Rayleigh scattering by H_2 due to the deeper homopause specific to Uranus, and methane absorption below the homopause. These processes have to be modeled simultaneously to determine the final intensity of the Lyman α emission at Uranus corresponding to each HST observation date.

Uranus orbits ~ 19 AU from the Sun, where both the Sun and IPH are sources of Lyman- α photons (Herbert & Hall, 1996; Yelle & Sandel, 1986). Resonant scattering by IPH also attenuates the solar Lyman- α flux en-route to Uranus, so both emission sources and IPH attenuation must be included in the modeling.

Uranus' atmospheric Lyman- α emission is optically thick (Ben Jaffel et al., 1991; Joshi et al., 2025), requiring a radiative-transfer (RT) model that treats multiple scattering. We employ a doubling-and-adding RT approach (Stephenson et al., 2024; Yelle, 1988) assuming a plane-parallel atmosphere with multiple layers. Within each layer, resonant scattering of solar and IPH Lyman- α photons by atomic H, Rayleigh scattering by H_2 , and absorption by CH_4 are included. The model also accounts for wing scattering in the H Lyman- α line profile via the angle-dependent partial-frequency-redistribution method (Bush & Chakrabarti, 1995; Hummer, 1962), treating H, H_2 , and CH_4 self-consistently throughout the atmosphere.

Solar and IPH Lyman- α photons are treated as distinct sources. Solar photons enter Uranus' atmosphere from a single direction, whereas IPH photons arrive isotropically from 2π steradian of sky. The IPH component accounts for directional variations in the Lyman- α line profile, which depend on the temperature and intensity of each line-of-sight column (Quémerais et al., 2013). Both the solar Lyman- α flux and the IPH sky brightness at Uranus vary

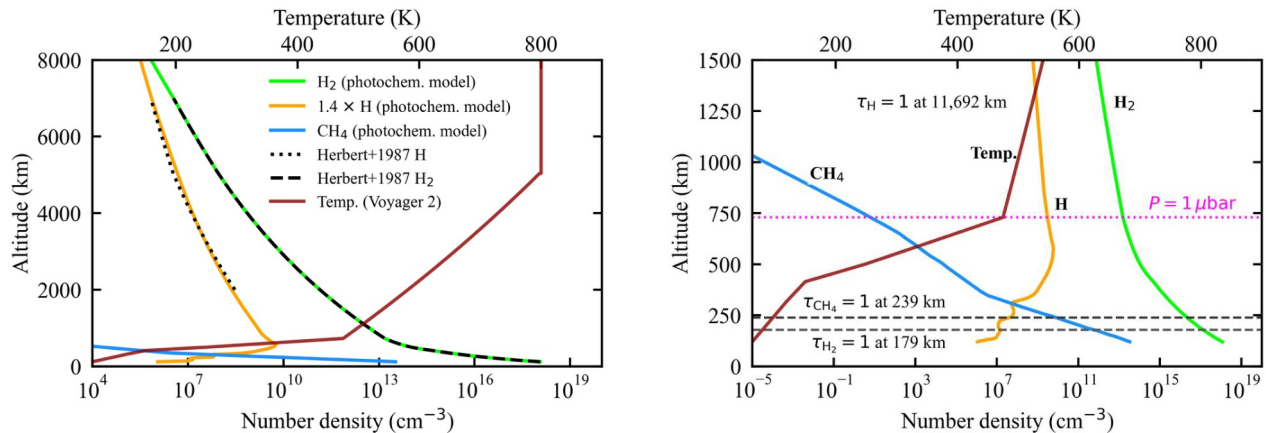


Figure 1. (a) Atmospheric structure from Voyager 2 data used to model Uranus' Hubble Space Telescope Lyman α emission. (b) Expanded view showing the 1-bar pressure level and altitudes where $\tau = 1$ for key species.

with solar activity on the date of the HST observation, and these variations are included in the modeling (Table S1 in Supporting Information S1).

The solar Lyman- α line-integrated flux for each HST observation date is obtained from the Laboratory for Atmospheric and Space Physics (LASP) Interactive Solar Irradiance Datacenter (LISIRD) (Lindholm & Wilson, 2013). To account for the difference in solar longitude between Earth and Uranus, the LISIRD flux is taken for the day when the same solar hemisphere faced Uranus as on the HST observation date. The resulting flux is used to determine the solar Lyman- α line shape and intensity at 1 AU (Emerich et al., 2005), which are then scaled to Uranus' heliocentric distance and corrected for wavelength-dependent attenuation by IPH using the model of Quémerais et al. (2013). The IPH-modified line shape is unique to each HST observation (see Figure S5 in Supporting Information S1).

Because HST observes from low Earth orbit, the modeled Uranus exospheric Lyman- α intensity must be corrected for attenuation by IPH column between Uranus and Earth. This correction depends on the wavelength-dependent IPH optical-depth profile and the relative motion between Uranus and the IPH flow direction. The effect is specific to each observation date (see Figure S4 in Supporting Information S1). The model incorporates this correction to yield the final Lyman- α intensity observed by HST.

An additional effect to be modeled is absorption of the Uranus Lyman α photons by the geocorona. Because the Uranus Lyman α line is broadened to ~ 1 Å by Rayleigh scattering from H₂, the geocoronal line cannot be completely Doppler shifted out of the planetary profile. A conservative upper limit was therefore adapted by assuming saturated absorption ($\tau \rightarrow \infty$) across the full width half max of a 1,000K Gaussian geocoronal profile, consistent with typical geocoronal temperatures (Baliukin et al., 2019; Connor et al., 2024). The true absorption is likely smaller, but a quantitative estimate would require knowledge of the hydrogen column density along the HST line of sight, which varies continuously over the course of a STIS exposure. This is further complicated by the poorly constrained nightside structure of the geocorona (Hodges, 1994; Kameda et al., 2017; Zoenchen et al., 2015). In practice, geocoronal absorption is expected to reduce the Uranus Lyman- α signal by only $\sim 3\%$ – 5% , and in some observing geometries by as little as $\sim 1\%$, due to the relative Doppler shift between Earth and Uranus (see Figure S4 in Supporting Information S1).

The RT model requires density profiles for H, H₂, and CH₄ as well as the temperature profile for Uranus to compute Uranus' Lyman α intensity. Our model's atmospheric structure (Figure 1) is anchored to the atmosphere retrieved from Voyager 2 UVS solar and stellar occultations (Herbert et al., 1987). The Voyager 2 derived atomic H density profile is well constrained at altitudes above 2,000 km, but the profile shape is unknown deeper in the atmosphere. Therefore, below 2,000 km, we obtain the vertical profiles for H, H₂ and CH₄ from the JPL/KINETICS photochemical model of Uranus which includes coupled ion-neutral chemistry for 321 species and over 2,500 reactions, solving the continuity equations for all species (Brown et al., 2024; Moses et al., 2020, 2023). However, this model underpredicts atomic H densities by a factor of ~ 1.4 when compared to the Voyager 2

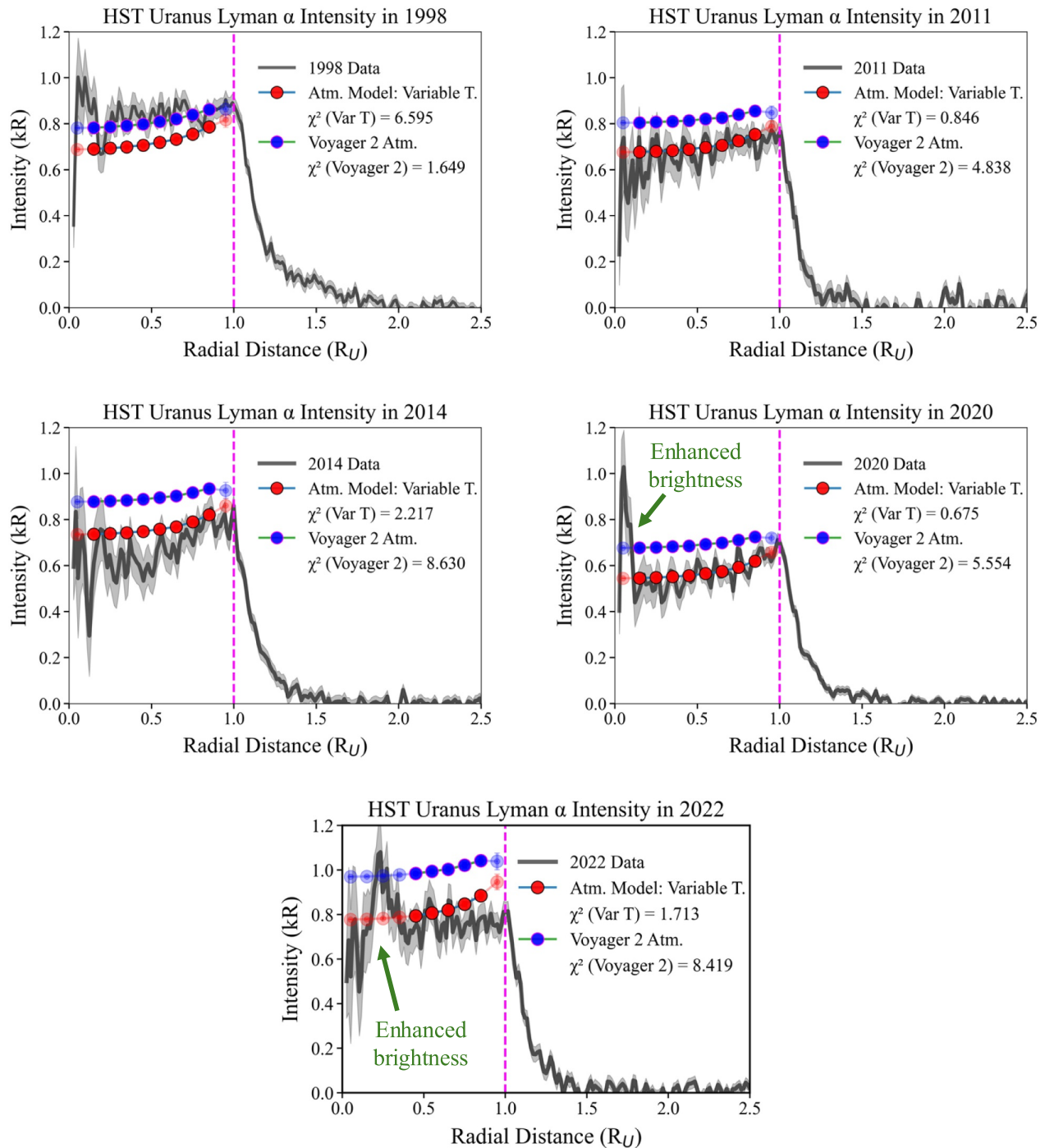


Figure 2. Modeled Uranus Lyman α disk intensity using the Voyager 2 atm (blue) and the same profile with reduced H scale height from H_3^+ temperatures (red). The red fit improves after 1998, indicating an exospheric scale height decrease consistent with the thermospheric cooling trend (Melin et al., 2019). Gray shaded region represents the uncertainty in the observed intensity. The lightly shaded blue and red model points represent points not considered in the χ^2 calculation to avoid modeling bias and the presence of auroral features in the data.

densities. Therefore, we scale the modeled atomic H profile by $1.4\times$ to match the Voyager 2 profile at higher altitudes. The temperature profile is the one derived from Voyager 2 observations of 1986.

The simulated Lyman α emission based on the Voyager 2 atmosphere model is compared against the HST observations in Figure 2 represented by the blue dotted line. The χ^2 of the fit is calculated using the expression

$\chi^2 = \frac{1}{n} \sum_{i=1}^n \frac{(I_{O,i} - I_{m,i})^2}{\sigma_{O,i}^2 + \sigma_{m,i}^2}$ where n is the number of model points represented by the dots in the model profile in

Figure 2, $I_{O,i}$ is the observed intensity with $\sigma_{O,i}$ the corresponding uncertainty for the i th point, and $I_{m,i}$ is the modeled intensity with $\sigma_{m,i}$ the corresponding model uncertainty for the i th point. The model uncertainties are represented by the standard deviation of Lyman- α brightness within each radial bin across the disk, reflecting spatial variations driven by differences in local insolation and the incident IPH flux. This is not a true representation of the actual model uncertainty which is much more complicated given the various assumptions used in the modeling process as well as the uncertainties in the various data sets used in constraining the models. Deriving the actual model uncertainty is beyond the scope of this work and will be explored in detail in a future publication.

The Voyager 2 atmosphere model matches the 1998 HST observation but overestimates Lyman α intensities for 2011–2022 despite accounting for observation geometry, IPH + geocorona conditions, solar activity, and Uranus–Sun distance. The discrepancy likely reflects that the fixed 1986 structure omits the thermospheric cooling seen in H_2 and H_3^+ data. In this work, we estimated this continued cooling of the thermosphere by reducing the scale height of the H column density by a factor consistent with the decrease in temperature. This is done by noting that the disk brightness is directly related to the vertical H column that can be crudely estimated as $\sim nh$ where n is the reference number density at the bottom of the column and h is the density scale height, and $h \propto T$. Thus the reduction in scale height is given by the relation: factor = $\frac{T_{\text{year}}}{T_{1992}}$ where T_{year} represents the thermospheric temperature determined from the fit to the H_3^+ emission data for a particular year as presented in Figure 4 of Melin et al. (2019) and T_{1992} represents the thermospheric temperature value on the fitted line corresponding to the first measurement of H_3^+ emission at Uranus (Trafton et al., 1999). The factors come out to be 0.9 for 1998, 0.76 for 2011, 0.73 for 2014, 0.66 for 2020, and 0.64 for 2022. For the determination of the factors, we assume that the atmosphere of Uranus has remained unchanged from 1986 (Voyager 2 flyby) to 1992 (first H_3^+ derived temperature). The same factor is not applied to the H_2 column density as it is optically thin at Lyman α and the major contribution for H_2 line-of-sight Rayleigh scattering intensity comes from below the 1 μ bar level where the temperature has remained unchanged since 1986 (Figure 1; Orton, Fletcher, et al., 2014; Orton, Moses, et al., 2014, Orton et al., 2015; see Figure S6 in Supporting Information S1). The H Lyman α emission though contains contributions from all altitudes, a significant portion of which comes from altitudes >1 μ bar pressure level which has cooled since 1986 (see Figure S6 in Supporting Information S1). This method is only used to model the Uranus disk brightness as it is based on disk-integrated H_3^+ derived temperatures applied to a plane-parallel radiative transfer model that cannot be applied to the limb emissions which would require time-dependent, altitude resolved temperature profiles that are unavailable. The resulting simulated Lyman α intensity represented by the red dots in Figure 2 matches the HST observed disk Lyman α intensity of Uranus for 2011, 2014, 2020, and 2022 better than the Voyager 2 atm Lyman α intensity (blue dots). This model does predict a lower intensity for 1998 than the results using the Voyager 2 derived atmosphere. The red dotted line fit for 2022 is slightly worse than the other years due to the presence of intense auroral activity away from the center of the planet and may therefore affect the true shape of the intensity curve of Uranus' disk Lyman α profile. The modeling results suggest that even though the thermospheric cooling started after the Voyager 2 flyby at the altitude range of the IR emissions indicated by the H_3^+ data, the cooling trend did not affect the full altitude range of scattering of Lyman α until after 1998.

The Lyman α intensity profiles for 2020 and 2022 contain a region of enhanced brightness as indicated in Figure 2. These regions correspond to auroral activity which has been previously identified in the same HST data set by Lamy et al. (2017, 2025). We find the average increased intensity to be ~ 467 Rayleighs for 2020 and ~ 326 Rayleighs for 2022 above the average disk intensity of ~ 562 Rayleighs for 2020 and ~ 754 Rayleighs for 2022. These values are consistent with previously observed values for H Lyman α + H_2 aurora at Uranus (Broadfoot et al., 1986; Herbert, 2009; Herbert & Sandel, 1994; Lamy et al., 2017). The presence of auroral activity does not appear to increase the overall temperature of the exosphere at Uranus.

Figure 3 shows the H column densities along HST's line of sight derived for the Uranus observations based on the variable temperature atmosphere model (red circles in Figure 2). We find that the H column in the model decreases with time indicating a steady collapse of the Uranian exosphere. The Voyager atmospheric H column, on the other hand, remains unchanged with time and doesn't match the HST observations well after 1998 (blue circles in Figure 2).

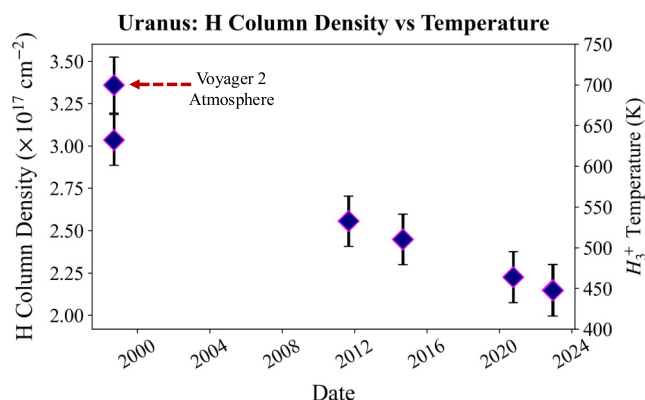


Figure 3. H column densities inferred from the model with variable temperature (red circles in Figure 2). The point marked as the “Voyager 2 Atmosphere” is the column density of H at Uranus in 1998 for Voyager 2 observation conditions.

Formal error bars shown in Figure 3 represent the propagated uncertainty in the inferred H column densities arising from the spread in thermospheric temperatures derived from the H_3^+ infrared observations (see Table S1 in Supporting Information S1). Because the modeled H column densities scale with the density scale height ($h \propto T$), the relative uncertainty in temperature directly translates into a proportional uncertainty in column density. These error bars therefore reflect the temperature-driven variability in the model but do not include additional systematic uncertainties associated with the Voyager 2 measurements or the photochemical modeling. Inclusion of such systematic uncertainties would increase the absolute error bars, but not to an extent that would alter the observed decrease in the H column density between the 1998 Voyager 2 value and 2022.

3. Discussion and Conclusion

Analysis of HST observations of Uranus' disk Lyman- α emission from 1998 to 2022 shows that the atmospheric altitudes responsible for Lyman- α scattering did not undergo significant cooling until after 1998 even though H_3^+ observations suggest that the thermosphere cooled by ~ 150 K from 1986 to

1998. Temperatures derived from the H_2 quadrupole line emissions from 1993 agree well with H_3^+ temperatures from 1992 to 1993 (Melin et al., 2019; Trafton & Kaplan, 2025). However, because H_2 temperatures probe a deeper pressure level than H_3^+ , it suggests that the lower thermosphere was hotter than that indicated by the H_3^+ cooling trend in 1993, if we assume that the upper atmospheric cooling started immediately after the Voyager 2 flyby in 1986. Therefore, there is some uncertainty with regards to the actual timeline of the upper atmospheric cooling of Uranus if we consider all three emissions: H_3^+ probing the upper thermosphere, H_2 the lower thermosphere, and H Lyman α a broad range of altitudes. Short-term temperature surges found in the H_3^+ emission data (Melin et al., 2019) may also explain the higher temperature required to model the 1998 HST observations. One might also consider the idea of the solar wind dynamic pressure driving the cooling trend at Uranus as proposed by Masters et al. (2024). However, because solar wind dynamic pressure partly correlates with the solar cycle, this challenges the idea that the cooling trend is driven by the change in incident solar wind dynamic pressure on the Uranus system. A similar argument was put forward in Trafton and Kaplan (2025) who pointed out that the solar wind dynamic pressure has not exhibited a steady decrease since 2010 while the cooling trend has continued. We also note that the H_3^+ and H_2 temperatures derived since the 1990s exhibit no obvious correlation with the peaks in incident solar wind dynamic pressure that occur during solar maxima. At present, what remains to be seen is if the cooling trend bears any correlation with the magnetic season at Uranus and if the much-awaited temperature reversal happens at the turn of the magnetic season after April 2030.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The Hubble Space Telescope observations used in this analysis is publicly available through the HST MAST archive at <https://mast.stsci.edu/search/ui/#/hst>. The data and software for figures are available through Zenodo at Bhattacharyya (2025).

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